# Lecture 29

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## Contents

**1 Newton Sums and Symmetric Polynomials** **1**
1.1 Newton Sums for Quadratics . . . . . 1
1.2 Cubic Polynomials and Vieta's Formulas . . . . . 2
1.2.1 Vieta's Formulas . . . . . 2
1.2.2 Asymptotic Behavior . . . . . 2

**2 Solving Polynomial Equations** **2**
2.1 Cubic Equations (Cardano's Formula) . . . . . 2
2.2 Higher Degree Polynomials . . . . . 3

**3 Problems** **3**
3.1 Problem 1: Constructing Polynomials . . . . . 3
3.2 Problem 2: System of Power Sums . . . . . 3
3.3 Problem 3: Polynomial Identity . . . . . 4

**4 Lagrange Interpolation** **4**
4.1 Problem Statement . . . . . 4
4.2 Construction . . . . . 4
4.3 General Formula . . . . . 5

# 1 Newton Sums and Symmetric Polynomials

## 1.1 Newton Sums for Quadratics

Consider the polynomial $ax^2 + bx + c$ with roots $x_1, x_2$. Let $S_n = x_1^n + x_2^n$. The recurrence relation for the sums of powers is given by:

$$aS_n + bS_{n-1} + cS_{n-2} = 0$$

### 1.2 Cubic Polynomials and Vieta's Formulas

Consider a cubic polynomial $p(x) = x^3 + ax^2 + bx + c$. Let the roots be $x_1, x_2, x_3$. Any symmetric polynomial in the roots, $Q(x_1, x_2, x_3)$, remains unchanged under any permutation of variables (e.g., $Q(x_2, x_1, x_3) = Q(x_1, x_2, x_3)$).

#### 1.2.1 Vieta's Formulas

By expanding the factored form $p(x) = (x - x_1)(x - x_2)(x - x_3)$:

$$p(x) = x^3 - (x_1 + x_2 + x_3)x^2 + (x_1x_2 + x_2x_3 + x_1x_3)x - x_1x_2x_3$$

Comparing coefficients with $x^3 + ax^2 + bx + c$:

1. $x_1 + x_2 + x_3 = -a$
2. $x_1x_2 + x_2x_3 + x_1x_3 = b$
3. $x_1x_2x_3 = -c$

#### 1.2.2 Asymptotic Behavior

- If the leading coefficient is positive:
  - As $x \to +\infty$, $p(x) \to +\infty$.
  - As $x \to -\infty$, $p(x) \to -\infty$.
- A cubic polynomial with real coefficients must have at least one real root (Intermediate Value Theorem).

## 2 Solving Polynomial Equations

### 2.1 Cubic Equations (Cardano's Formula)

For the depressed cubic $y^3 + py + q = 0$, the roots are given by:

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{R}} + \sqrt[3]{-\frac{q}{2} - \sqrt{R}}$$

where the discriminant is $R = \left(\frac{p}{3}\right)^3 + \left(\frac{q}{2}\right)^2$. **Note:** General cubics $x^3 + ax^2 + bx + c = 0$ are converted to depressed form via the substitution $x = y - \frac{a}{3}$.

### 2.2 Higher Degree Polynomials

- **Quartic (Degree 4):** Solvable by radicals (Ferrari's method).
- **Quintic (Degree 5) and higher:** No general formula involving only arithmetic operations and radicals exists (Abel-Ruffini Theorem / Galois Theory, c. 1830).

## 3 Problems

### 3.1 Problem 1: Constructing Polynomials

Find a polynomial in $\mathbb{Z}[x]$ with root $\sqrt{3}+2$. **Solution:** Let $x = 2+\sqrt{3}$. Since coefficients are integers, the conjugate $2-\sqrt{3}$ must also be a root.

$$p(x) = (x-(2+\sqrt{3}))(x-(2-\sqrt{3}))$$

$$p(x) = ((x-2)-\sqrt{3})((x-2)+\sqrt{3})$$

$$p(x) = (x-2)^2 - 3 = x^2 - 4x + 4 - 3 = x^2 - 4x + 1$$

### 3.2 Problem 2: System of Power Sums

Solve for $x, y, z$:

$$\begin{cases} x+y+z = 2 \\ x^2+y^2+z^2 = 14 \\ x^3+y^3+z^3 = 20 \end{cases}$$

**Solution:** Let $e_1, e_2, e_3$ be the elementary symmetric polynomials.

1. $e_1 = x+y+z = 2$.
2. From $(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx)$:

$$2^2 = 14 + 2e_2 \implies 4 = 14 + 2e_2 \implies 2e_2 = -10 \implies e_2 = -5$$

3. Using the algebraic identity relating power sums:

$$(x+y+z)(x^2+y^2+z^2-(xy+yz+zx)) + 3xyz = x^3+y^3+z^3$$

$$e_1(14-e_2) + 3e_3 = 20$$

$$2(14-(-5)) + 3e_3 = 20$$

$$2(19) + 3e_3 = 20 \implies 38 + 3e_3 = 20 \implies 3e_3 = -18 \implies e_3 = -6$$

The variables $x, y, z$ are roots of the cubic equation:

$$t^3 - e_1 t^2 + e_2 t - e_3 = 0$$

$$t^3 - 2t^2 - 5t + 6 = 0$$

By inspection, $t_1 = 1$ is a root ($1 - 2 - 5 + 6 = 0$). Dividing by $(t - 1)$:

$$t^3 - 2t^2 - 5t + 6 = (t-1)(t^2 - t - 6) = (t-1)(t-3)(t+2)$$

The roots are $1, 3, -2$. **Answer:** The set $\{x, y, z\}$ is $\{1, 3, -2\}$.

### 3.3 Problem 3: Polynomial Identity

Solve for $x$:

$$c\frac{(x-a)(x-b)}{(c-a)(c-b)} + b\frac{(x-a)(x-c)}{(b-a)(b-c)} + a\frac{(x-b)(x-c)}{(a-b)(a-c)} = x$$

**Solution:** Define $P(x)$ as the LHS. $P(x)$ is a polynomial of degree $\leq 2$. Let $Q(x) = P(x) - x$. The degree of $Q(x)$ is $\leq 2$. Check specific values:

- At $x = c$: $P(c) = c \cdot 1 + 0 + 0 = c$. So $Q(c) = 0$.
- At $x = b$: $P(b) = 0 + b \cdot 1 + 0 = b$. So $Q(b) = 0$.
- At $x = a$: $P(a) = 0 + 0 + a \cdot 1 = a$. So $Q(a) = 0$.

Since $Q(x)$ has 3 distinct roots $(a, b, c)$ but degree $\leq 2$, it must be the zero polynomial. Thus, the equation holds for all $x$.

## 4 Lagrange Interpolation

### 4.1 Problem Statement

Given $n+1$ distinct points $(x_1, y_1), (x_2, y_2), \ldots, (x_{n+1}, y_{n+1})$, is there a polynomial that passes through these points? i.e., $p(x_i) = y_i$.

### 4.2 Construction

We construct basis polynomials that are 1 at a specific node and 0 at all others. For the first point $(x_1, y_1)$:

$$p_1(x) = y_1 \cdot \frac{(x - x_2)(x - x_3)\ldots(x - x_{n+1})}{(x_1 - x_2)(x_1 - x_3)\ldots(x_1 - x_{n+1})}$$

Observe that:

- $p_1(x_1) = y_1$
- $p_1(x_k) = 0$ for $k \neq 1$.

### 4.3 General Formula

The interpolating polynomial is the sum of these basis polynomials:

$$P(x) = \sum_{i=1}^{n+1} p_i(x) = \sum_{i=1}^{n+1} y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

This polynomial satisfies $P(x_k) = y_k$ for all $k$, and $\deg(P) \leq n$.